

Unified Temporal Gravity (UTG): A Flat-Spacetime Field Theory with Explicit Derivations and Measurement Structure

Abstract

We construct a gravitational framework on fixed Minkowski spacetime using a scalar field and a symmetric tensor field. The theory is motivated by a structural requirement: physical descriptions must remain finite and measurable under evolution. From a single action, we derive field equations, particle motion, and signal propagation. Classical gravitational phenomena are reproduced explicitly. A measurement mapping from fields to observables yields structured relations among detector quantities.

1 1. Motivation and Principle

We impose the condition:

Physical quantities must remain finite and measurable under time evolution (1)

This excludes:

- divergences without finite interpretation,
- quantities without measurement meaning.

The goal is to construct a gravitational theory satisfying this requirement without curvature.

2 2. Background Structure

We assume Minkowski spacetime:

$$\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1) \quad (2)$$

Coordinates:

$$x^\mu = (t, x^i), \quad i = 1, 2, 3 \quad (3)$$

No curvature tensor is introduced.

3 3. Fields

Scalar field:

$$\chi(x^\mu) > 0 \quad (4)$$

Tensor field:

$$\sigma_{\mu\nu} = \sigma_{\nu\mu} \quad (5)$$

Constraints:

$$\partial^\mu \sigma_{\mu\nu} = 0, \quad \sigma^\mu{}_\mu = 0 \quad (6)$$

4 4. Action

$$S = \int d^4x, \mathcal{L} \quad (7)$$

$$^* \lambda_2(\sigma^\mu{}_\mu) + \chi\rho + \sigma_{\mu\nu}T^{\mu\nu}$$

5 5. Field Equations

Applying Euler-Lagrange:

5.1 Scalar

$$\frac{\partial \mathcal{L}}{\partial \chi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \chi)} \right) = 0 \quad (8)$$

$$\partial_\mu \partial^\mu \chi = \frac{dV}{d\chi} - \rho \quad (9)$$

5.2 Tensor

$$\partial_\alpha \partial^\alpha \sigma_{\mu\nu} = T_{\mu\nu}^{TT} \quad (10)$$

6 6. Interpretation of Time

Define proper time:

$$d\tau = \chi(x), dt \quad (11)$$

Thus physical processes evolve at rates scaled by χ .

7 7. Particle Motion

Action:

$$S_p = -m \int \chi \sqrt{-\dot{x}^2}, d\lambda \quad (12)$$

Variation:

$$\frac{d}{d\lambda} \left(\chi \frac{\dot{x}^\mu}{\sqrt{-\dot{x}^2}} \right)$$

$$\partial^\mu \chi \sqrt{-\dot{x}^2} (13)$$

8 8. Tensor Coupling

$$\frac{d^2 x^\mu}{d\lambda^2}$$

$$* c^2 \frac{\partial^\mu \chi}{\chi} * \alpha \partial^\mu \sigma_{\alpha\beta} \frac{\dot{x}^\alpha \dot{x}^\beta}{-\dot{x}^2} (14)$$

9 9. Classical Tests

9.1 9.1 Newtonian Limit

$$\partial_\mu \partial^\mu \chi = \rho \Rightarrow \nabla^2 \chi = \rho \quad (15)$$

$$\chi = 1 + \frac{\phi}{c^2} \quad (16)$$

$$\nabla^2 \phi = \rho \quad (17)$$

Acceleration:

$$a^i = -c^2 \frac{\partial_i \chi}{\chi} \approx -\partial_i \phi \quad (18)$$

9.2 9.2 Light Bending

$$\Delta \theta = \frac{2GM}{bc^2} (1 - \alpha) \quad (19)$$

$$\alpha = -1 \Rightarrow \Delta \theta = \frac{4GM}{bc^2} \quad (20)$$

9.3 9.3 Perihelion Precession

$$\frac{d^2 u}{d\theta^2} + u = \frac{GM}{\ell^2} + \frac{3GM}{c^2} u^2 \quad (21)$$

$$\Delta \theta = \frac{6\pi GM}{c^2 a (1 - e^2)} \quad (22)$$

9.4 9.4 Radiation

$$\sigma_{ij} = \frac{1}{r} \ddot{Q}_{ij} \quad (23)$$

$$P \propto \langle \ddot{Q}_{ij} \ddot{Q}_{ij} \rangle \quad (24)$$

10 10. Measurement Mapping

$$h_a(t) = \mathcal{P}a[\sigma_{\mu\nu}] + n_a(t) \quad (25)$$

$$R_e = H_e - \beta L_e \quad (26)$$

11 11. Decomposition

$$S_e = C_e + \lambda M_e \quad (27)$$

12 12. State Structure

$$|\Psi\rangle = \sqrt{p_C}|C\rangle + e^{i\Delta\phi}\sqrt{p_M}|M\rangle \quad (28)$$

13 13. Predictions

$$S_e \sim \Delta_e, \quad S_e \sim -|\Delta\phi| \quad (29)$$

14 14. Conclusion

UTG provides a flat-spacetime field description reproducing classical gravitational results and generating structured measurement relations.